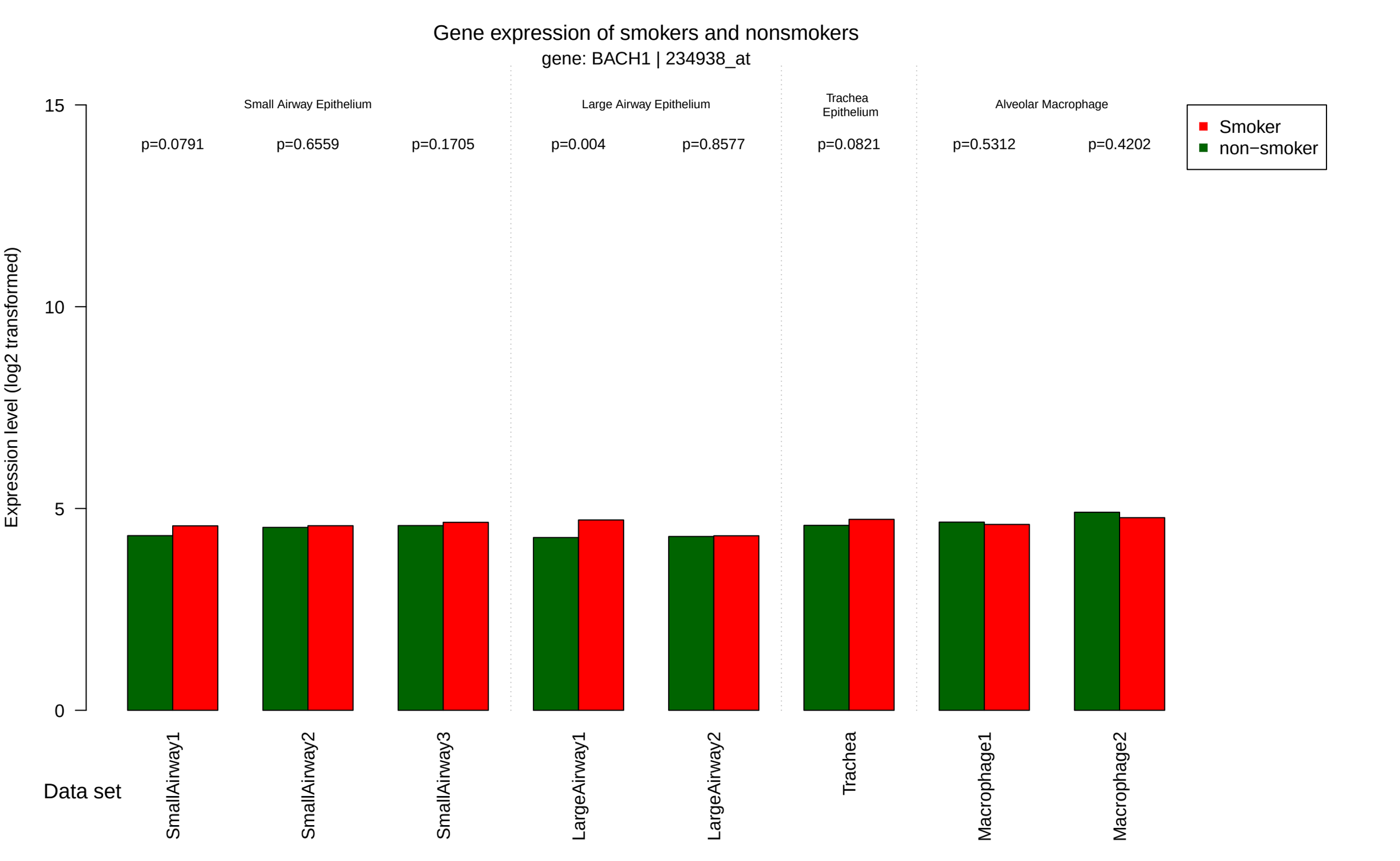
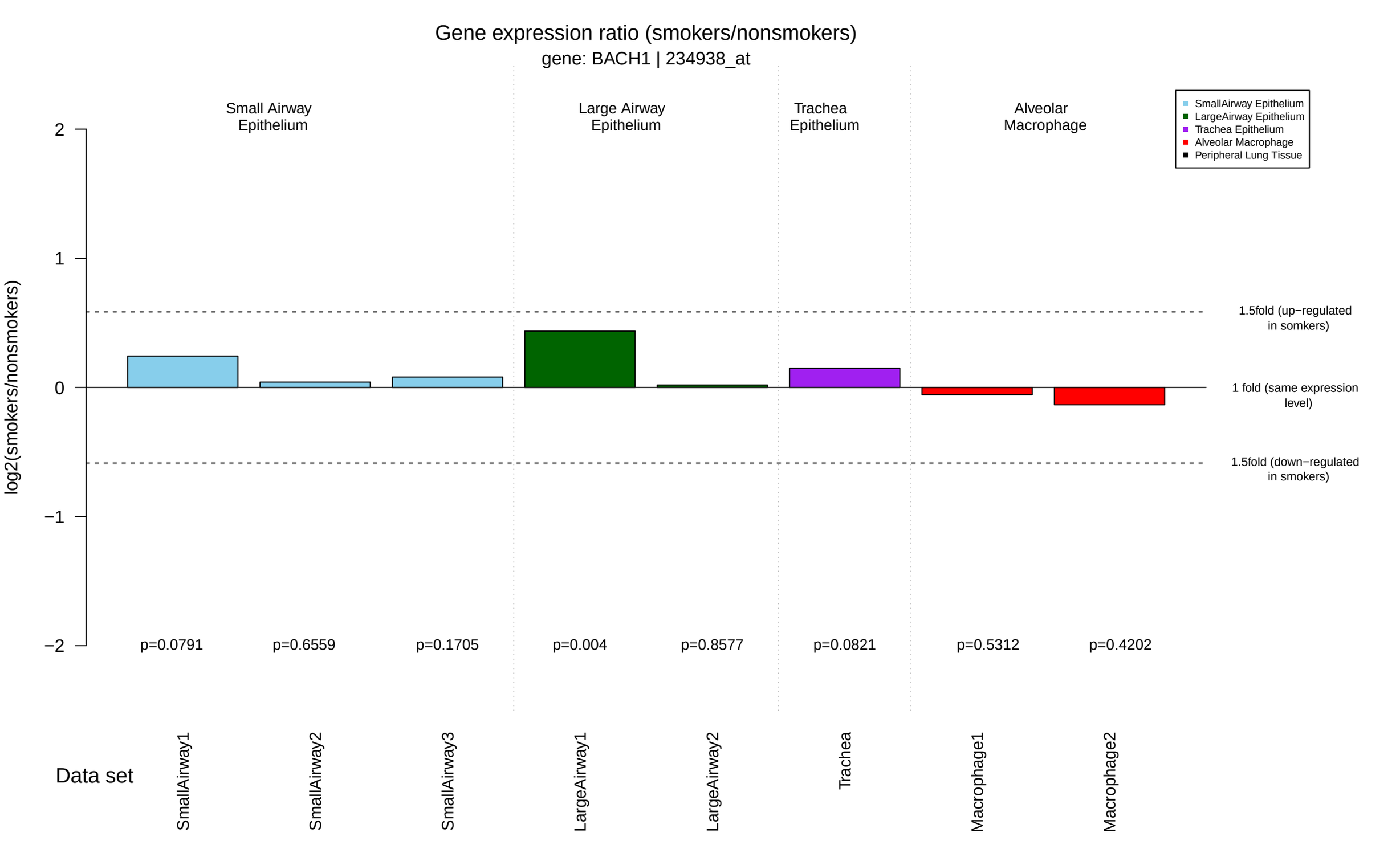
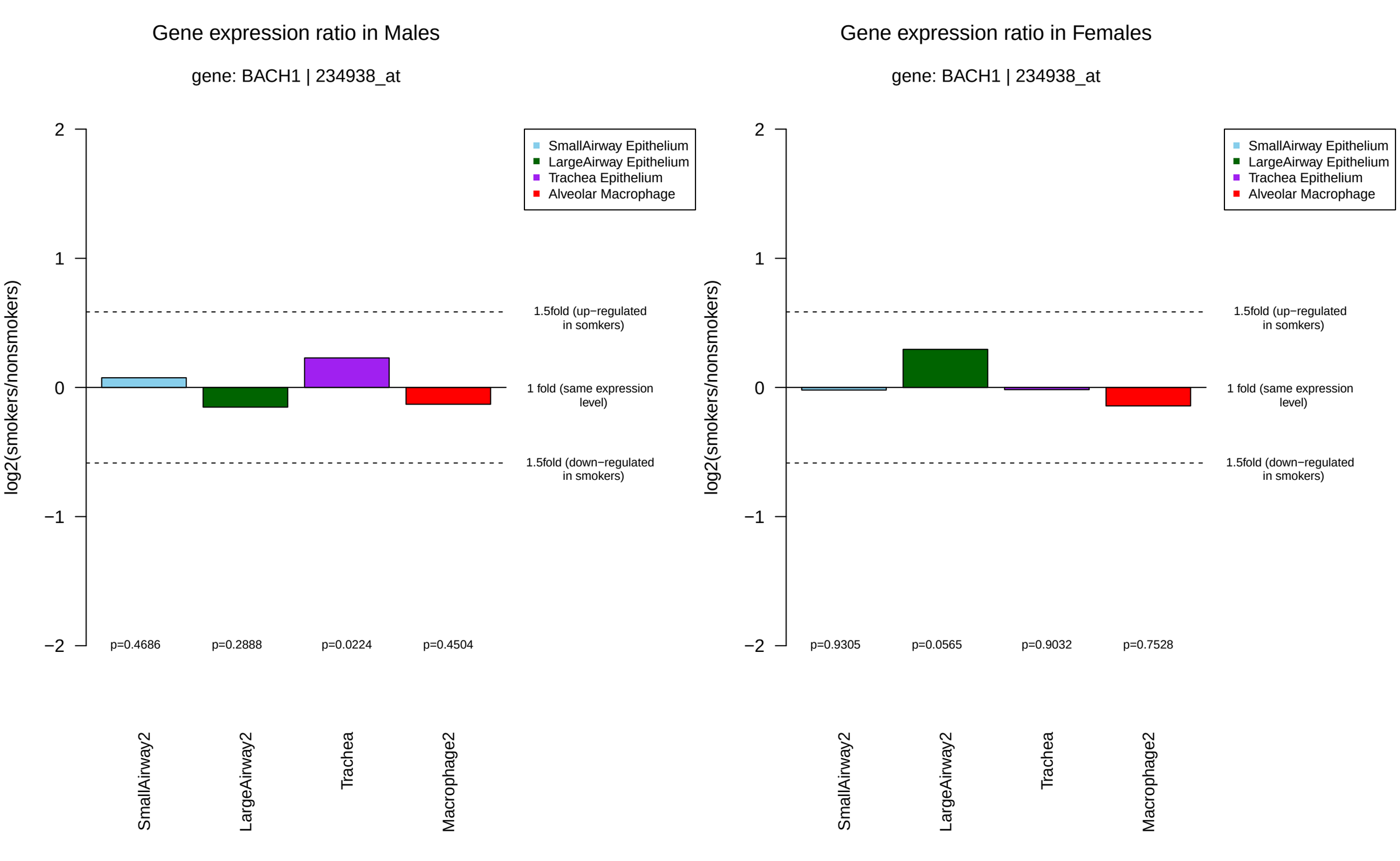
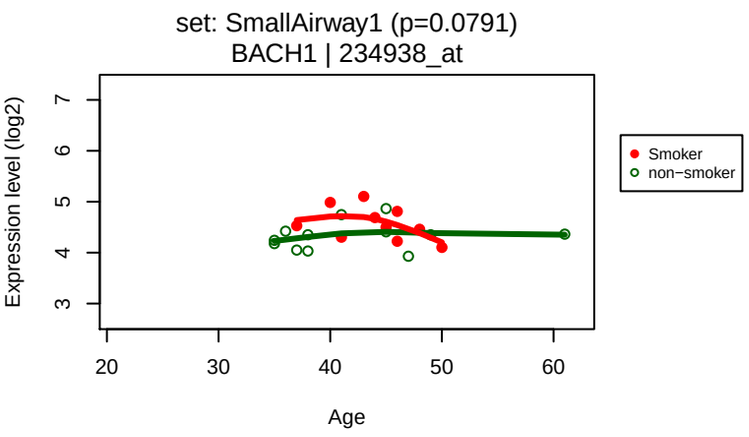
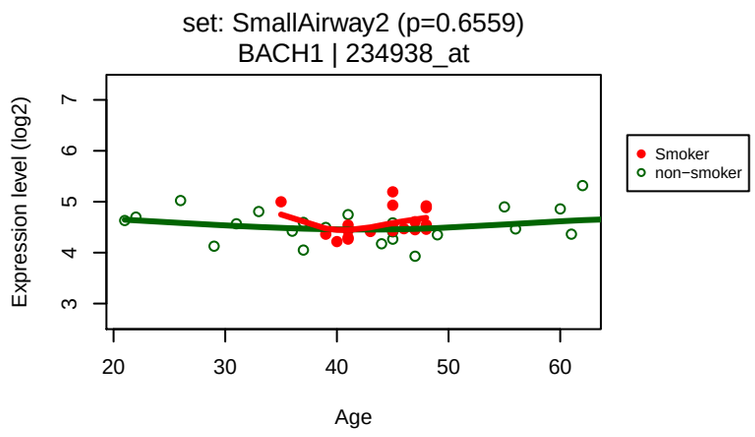
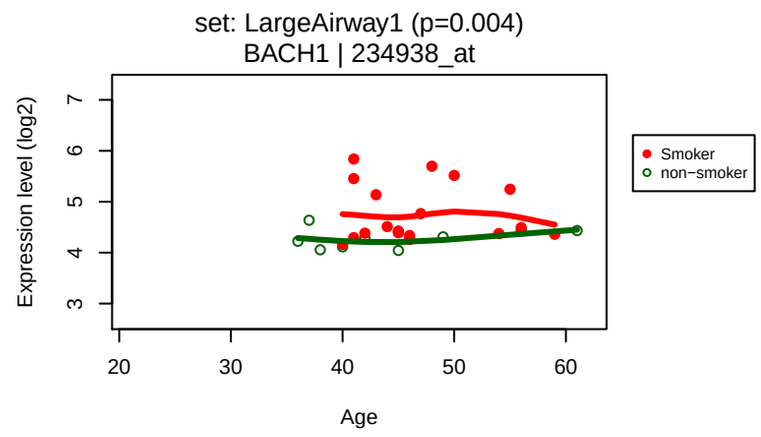

Gene expression of smokers and nonsmokers
gene: BACH1 | 234938_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0791
p=0.6559
p=0.1705
p=0.004
p=0.8577
p=0.0821
p=0.5312
p=0.4202
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

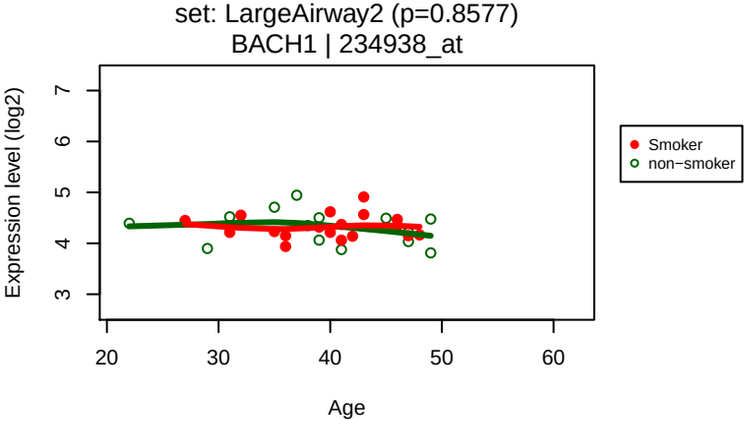

Gene expression ratio (smokers/nonsmokers)
gene: BACH1 | 234938_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0791
p=0.6559
p=0.1705
p=0.004
p=0.8577
p=0.0821
p=0.5312
p=0.4202
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

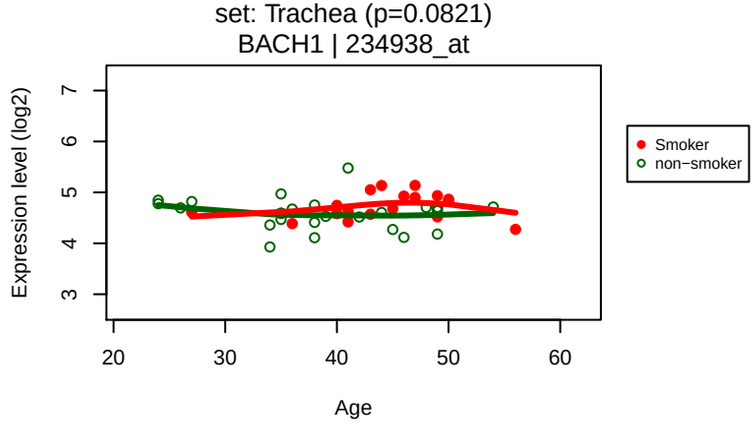

Gene expression ratio in Males
gene: BACH1 | 234938_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4686
p=0.2888
p=0.0224
p=0.4504
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BACH1 | 234938_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9305
p=0.0565
p=0.9032
p=0.7528
SmallAirway2
LargeAirway2
Trachea
Macrophage2

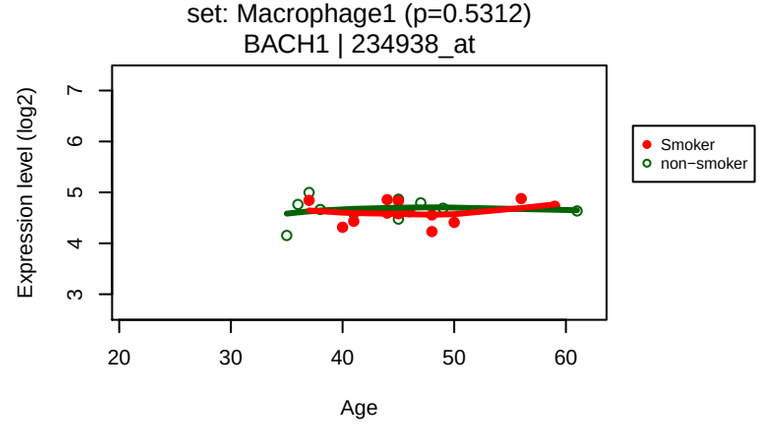

set: SmallAirway1 (p=0.0791)
BACH1 | 234938_at
Expression level (log2)
Age
Smoker
non−smoker

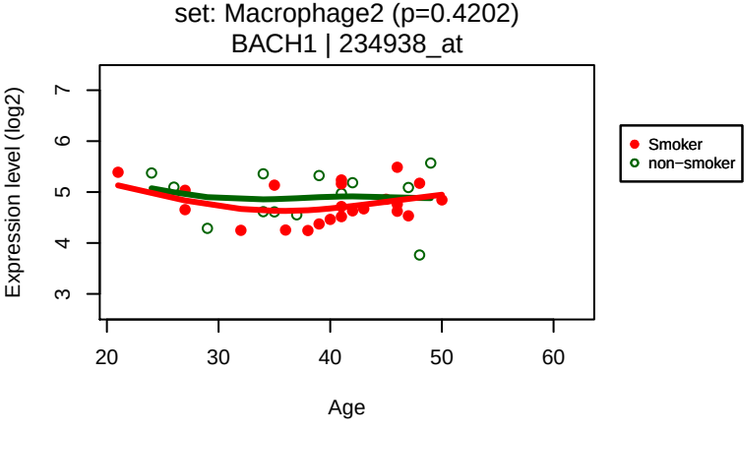

set: SmallAirway2 (p=0.6559)
BACH1 | 234938_at
Expression level (log2)
Age
Smoker
non−smoker

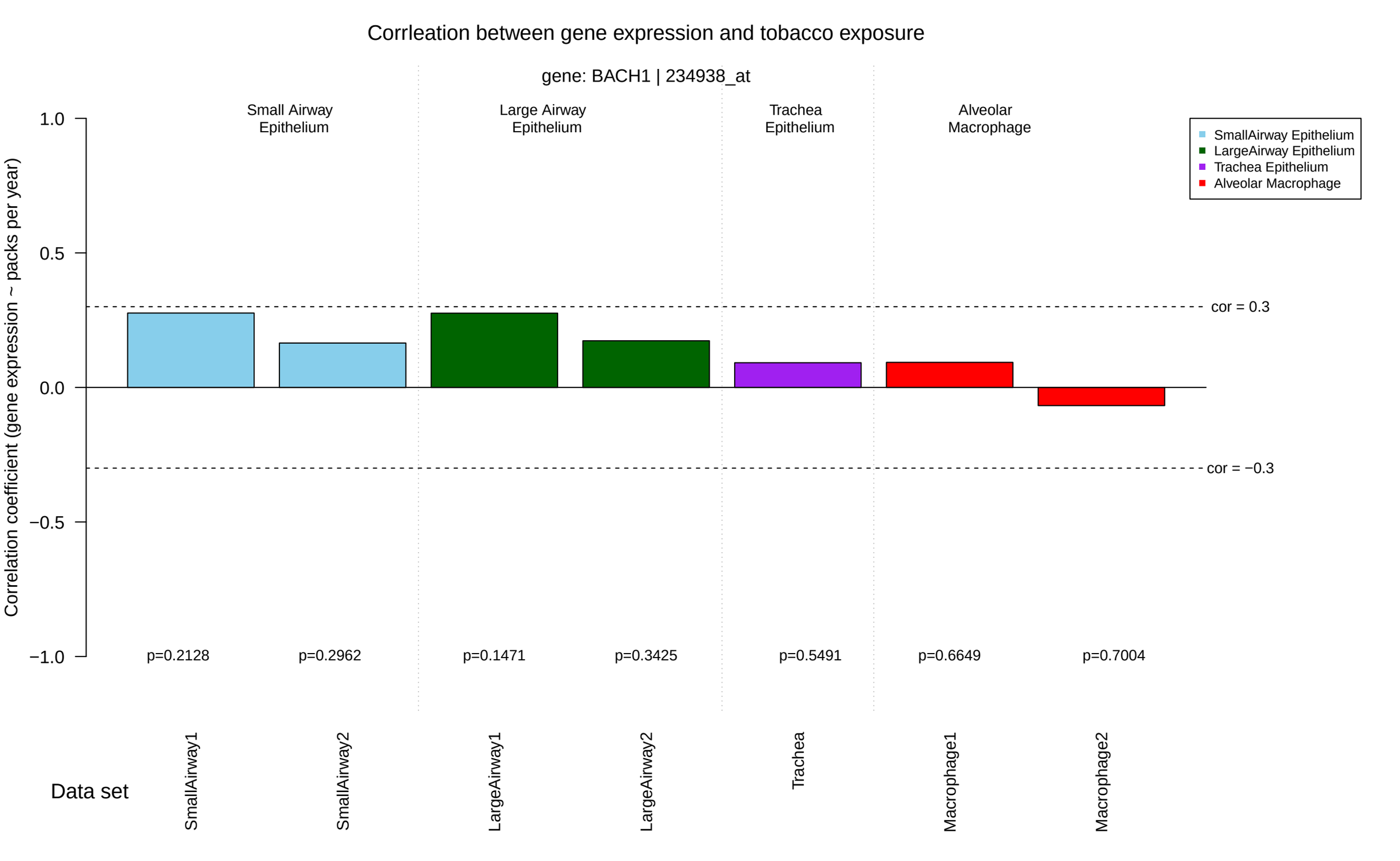

set: LargeAirway1 (p=0.004)
BACH1 | 234938_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8577)
BACH1 | 234938_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0821)
BACH1 | 234938_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5312)
BACH1 | 234938_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4202)
BACH1 | 234938_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BACH1 | 234938_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2128
p=0.2962
p=0.1471
p=0.3425
p=0.5491
p=0.6649
p=0.7004
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2